Problem 3: You are given the value of n. Calculate the result of $1 \times n + 2 \times (n-1) + \cdots + n \times 1$. (Mahbubul. 32)

Solution:

$$1 \times n + 2 \times (n-1) + 3 \times (n-2) + 4 \times (n-3) + \cdots n \times 1$$

$$\rightarrow n + 2n - 2 + 3n - 6 + 4n - 12 + \cdots n \times (n-(n-1))$$

$$\rightarrow (1n + 2n + 3n + 4n + \cdots n \times n) - (2 + 6 + 12 + \cdots n(n-1))$$

$$\rightarrow n(1 + 2 + 3 + 4 + \cdots n) - 2\left(1 + 3 + 6 + \cdots \frac{n(n-1)}{2}\right)$$

$$\rightarrow n \sum_{i=1}^{n} (i) - 2\left(\sum_{i=1}^{1} (i) + \sum_{i=1}^{2} (i) + \sum_{i=1}^{3} (i) + \cdots + \sum_{i=1}^{n-1} (i)\right)$$

Let, $n - 1 = z$

$$\rightarrow n \sum_{i=1}^{n} (i) - 2\left(\sum_{i=1}^{1} (i) + \sum_{i=1}^{2} (i) + \sum_{i=1}^{3} (i) + \cdots + \sum_{i=1}^{z} (i)\right)$$

$$\rightarrow n \sum_{i=1}^{n} (i) - 2 \sum_{j=1}^{z} \left(\sum_{i=1}^{j} (i)\right) \quad \ldots (a)$$

Now, $\sum_{j=1}^{z} \left(\sum_{i=1}^{j} (i)\right) = \sum_{j=1}^{z} \left(\frac{j(j+1)}{2}\right)$

$$= \sum_{j=1}^{z} \left(\frac{1}{2} j^2 + \frac{1}{2} j\right)$$

$$= \sum_{j=1}^{z} \left(\frac{1}{2} j^2\right) + \sum_{j=1}^{z} \left(\frac{1}{2} j\right)$$

$$= \frac{1}{2}\sum_{j=1}^{z}(j^2) + \frac{1}{2}\sum_{j=1}^{z}(j)$$

$$= \frac{1}{2}\left(\left(\frac{z(z+1)(2z+1)}{6}\right) + \left(\frac{z(z+1)}{2}\right)\right)$$

$$= \frac{1}{2}\left(\left(\frac{(n-1)((n-1)+1)(2(n-1)+1)}{6}\right) + \left(\frac{(n-1)((n-1)+1)}{2}\right)\right)$$

$$= \frac{1}{2}\left(\left(\frac{(n^2-n)(2n-1)}{6}\right) + \left(\frac{n^2-n}{2}\right)\right)$$

$$= \frac{1}{2}\left(\frac{2n^3 - \cancel{n^2} - \cancel{2n^2} + n + \cancel{3n^2} - 3n}{6}\right)$$

$$= \frac{1}{2}\left(\frac{2n^3-2n}{6}\right) \quad \ldots (b)$$

Back to (a),

$$n\sum_{i=1}^{n}(i) - 2\sum_{j=1}^{z}\left(\sum_{i=1}^{j}(i)\right)$$

$$\rightarrow \left(n\,\frac{n(n-1)}{2}\right) - \cancel{2} \times \frac{1}{\cancel{2}}\left(\frac{2n^3-2n}{6}\right)$$

$$\rightarrow \frac{n^3+n^2}{2} - \frac{2n^3-2n}{6}$$

$$\rightarrow \frac{\cancel{3}n^3 + 3n^2 - \cancel{2n^3} + 2n}{6}$$

$$\rightarrow \frac{n^3+3n^2+2n}{6}$$

Pre-assumptions:

1. Summation rules for closed-form expressions are known. Which are:

(a) $\sum_{k=1}^{n} (c) = nc$

(b) $\sum_{i=1}^{n} (c \cdot i) = c \sum_{i=1}^{n} (i)$

(c) $\sum_{i=1}^{n} (x_i \pm y_i) = \sum_{i=1}^{n} (x_i) + \sum_{i=1}^{n} (y_i)$

2. Summation formulas for close-form expressions are known. Which are:

(a) $\sum_{i=1}^{n} (i) = \frac{n(n+1)}{2}$

(b) $\sum_{i=1}^{n} (i^2) = \frac{n(n+1)(2n+1)}{6}$

3. There is no familiarity with pascal's triangle! We don't about it! So no shortcuts from there!

Pre-assumptions:

1. Same as problem 1.

Discussion:

Same as problem 1. Different expression of the same equation.

Observations:

1. $-(2+6+12+\cdots n(n-1))$ ← From this,

I made it to: $-2(1+3+6+\cdots+\frac{n(n-1)}{2})$

Here, I know, the minus terms will be divisible by 2 with no remainder. How?

the integers in $-(2+6+12+\cdots+n(n-1))$ are $i(i-1)$. So it does not matter if i is even or not. Because 2 integers that have 1 as the difference between them, 1 of them will always be even and even X (even or odd) = even.

Thus, $i(i-1)$ is always even.